ROADMAP

AI Agent Engineering Roadmap

A progressive learning path for engineers who build with LLMs — from prompt engineering to production multi-agent systems.

AI LITERATE — Understands AI concepts, uses AI tools effectively → **AI ENABLED ◀** — Builds apps and agents powered by LLMs → **AI NATIVE** — Trains and fine-tunes models from scratch

Covered in the AI Agent Engineering course · Explore on your own

The AI Agent Engineering course covers Phases 2–5. More courses coming soon.

1

LLM Foundations

- How LLMs work — tokens, context windows, temperature
- Prompt engineering — system prompts, few-shot, chain-of-thought
- LLM provider APIs — OpenAI, Anthropic, Google
- Structured output and function calling

2

RAG & Knowledge — MODULE 01

- Embeddings and vector databases — Pinecone, ChromaDB, Weaviate
- Chunking strategies
- RAG pipelines — retrieval-augmented generation
- Agentic RAG — the agent decides what to retrieve
- Reranking and hybrid search — keyword + semantic combined

3

Tool Use & Protocols — MODULE 03

- MCP — building and consuming tool servers
- Transport modes — Streamable HTTP and stdio
- Tool design patterns — what makes a good tool
- Client-server architecture for AI tools

4

AI Agents & Orchestration — MODULES 04 - 06

- The agentic loop — reason, act, observe
- LangGraph — state graphs, nodes, DAGs, fan-out/fan-in
- A2A protocol — distributed agents, agent cards, discovery
- Multi-agent patterns — parallel execution, orchestration

5

Local & Open-Source LLMs — MODULE 02

- Ollama — running models locally
- Model selection — size vs capability trade-offs
- When to use local vs cloud
- Open-source landscape — Llama, Mistral, Gemma, Qwen

BEYOND THE COURSE — KEEP GOING

6

Multimodal AI

- Vision — image understanding with GPT-4o, Claude, Gemini
- Image generation — DALL-E, Stable Diffusion, Flux
- Audio — speech-to-text (Whisper), text-to-speech (ElevenLabs)
- Building multimodal agents that combine all modalities
- Real-time voice — OpenAI Realtime API, conversational AI

7

Fine-Tuning & Customization

- When to fine-tune vs prompt engineer vs RAG
- HuggingFace — models, datasets, transformers library
- LoRA / QLoRA — efficient fine-tuning without massive compute
- Dataset preparation and evaluation

8

Production & Deployment

- Containerizing AI apps with Docker
- Cloud — AWS Bedrock, GCP Vertex AI, Azure AI
- Cost optimization — caching, model routing, token budgets
- Observability — LangSmith, tracing, logging agent runs
- Prompt caching — reducing cost and latency

9

Security & Safety

- Prompt injection — direct and indirect attacks and defenses
- Output validation, guardrails, and content moderation
- PII handling and data privacy
- Auth for AI services — who can call what

10

Evaluation & Testing

- Testing AI apps — it's not unit testing
- LLM-as-judge patterns
- Benchmark datasets and evaluation metrics
- Regression testing for prompt changes

Start your AI Agent Engineering journey — hands-on, from zero to multi-agent systems

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